

TOTAL SURVEY ERROR AND DATA QUALITY I
Survey Methodology 720 (SURV 720 / SURVMETH 720)
FALL 2025

Monday, 10:20 AM - 12:15 PM

University of Maryland - 1208 LeFrak Hall;
University of Michigan – 6080 ISR

Instructor: Sharan Sharma (snsharma@umd.edu)

Office Hours: No fixed hours; please email to set up a time to talk.

Overview of the Course

This course concerns the different sources of errors in estimates derived from survey data. It does not cover random sampling error or estimation but focuses on the other major sources of survey errors. More specifically, it concerns:

- 1) **Coverage error**, which results from the failure to give every member of the population a chance of selection into the sample
- 2) **Nonresponse error**, which results from the failure to collect data on all members of the sample
- 3) **Measurement error**, which results from the failure of the recorded responses to reflect the true characteristics of the respondents
- 4) **Editing, processing, and analysis errors**, which result from the failure to convert responses accurately into an analysis file and errors in analyzing the data themselves.

Apart from these, the course also discusses errors specifically associated with newer data sources.

The goal of survey design is to minimize the size of these and other errors (e.g., through interviewer training, sample design, or efforts at persuading sample persons to cooperate) subject to the cost constraints on any given survey. One difficulty in finding the best design reflects the fact that there are often tradeoffs between different sources of error. Each design feature also carries with it cost implications for the survey. In addition, several of these errors can be linked to one another in practice — attempting to decrease one may merely increase another (e.g., reducing nonresponse by aggressively persuading sample persons to cooperate may result in larger measurement errors in the survey data).

This course reviews research on these topics, first presenting the conceptual background behind each error source and then reviewing current research and gaps in the existing literature. It examines the interplay of errors and costs in survey designs. After introducing various conceptualizations of survey error, we will examine the different sources of survey error one by one. Although much of the survey methodology literature deals with one error source in isolation of others, we will integrate recent literature exploring the relationships among errors and discuss directions for future work.

This course presents research that attempts to examine the **causes** of survey errors. The course assumes that the students already know the basic steps of a survey research project. It is not a practicum in survey research, but instead covers many of the considerations on which survey design decisions should be based. This is not a "how-to" course, but rather investigates the basic principles, derived from the empirical literature, that might apply to diverse types of surveys.

The methods literature on survey error has two major strands, one exploring how to reduce survey errors and the other how to measure them. For each of the error sources, there will be readings on efforts to reduce the error and additional readings on how to measure them. In addition, we will also review theoretical perspectives on causes of errors. In short, for each error source we will address three questions:

1. What is the cause of the error?
2. What techniques can be used to reduce the error in practice?
3. What statistical models can be used to measure the error source?

Grading

Grading will be based on three components:

- Participation in class demonstrating understanding of the readings and in-class material (10% of grade);
- Weekly reading assignments (25% of grade);
- Midterm exam (30% of grade); and
- Final exam (35% of grade).

For the weekly graded reading assignments, you will annotate one of the assigned readings using a collaborative e-reader called *Perusall*. The goal is for you to share questions, answers, perspectives, and external knowledge in comment threads. You can annotate text, figures, or equations. The reading assignments are due by noon on Friday. The reading that you are asked to annotate each week is marked below with an asterisk (*). You can find the link to each reading assignment in the "Assignments" section of Canvas. This link will automatically direct you to the assigned reading in Perusall. Please be sure to annotate in Perusall and *not* using your own PDF reader.

Each reading assignment will be graded on a scale of 0-3. Your score is assigned based on the following components:

- annotation quality (thoughtful reading and interpretation)
- quantity (minimum 3 annotations per reading) and
- distribution (annotations not clustered in one part of the reading).

For more information about using Perusall, please see <https://support.perusall.com/hc/en-us/categories/360002173133-Students>.

The midterm and final examinations in the past have been open book, open notes and

administered via Canvas. However, I am starting to get concerned about students using AI, which is not allowed and destroys I will discuss this with you in class.

Resources and Readings

There is one required text for the course (Groves, R.M., *Survey Errors and Survey Costs*, Wiley, 2004). Note that the original version of this book published in 1989 is the same as the reprint published in 2004. Other required course readings will be available via Canvas. The list of readings provided below is subject to change, and students will be notified if there are any changes to the required readings on a weekly basis.

The list of readings may seem overwhelming but in our first class we will talk of reading strategies. Also see this useful paper:

Carey, M. A., Steiner, K. L., & Petri Jr, W. A. (2020). Ten simple rules for reading a scientific paper. *PLoS computational biology*, 16(7), e1008032. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7392212/>

Attending Class

Follow all university guidelines (UMD/U-M) when attending class.

If you are unable to attend a particular class in person, you can attend via Zoom with the instructor's permission. If extenuating circumstances prevent your attending a class at all, it is the student's responsibility to alert the instructor and make an alternative arrangement. If you have been approved to attend remotely, you can join the recurring Zoom "meeting" for Surv/Surv meth 720 (<https://umich.zoom.us/j/91990311672>); passcode: 7202025).

Audio and video will be recorded for each class session. These recordings are to be used to review lectures, or on those rare occasions when a student must miss class, to watch the class at another time. They are available on the Canvas course site under the Zoom page in the Cloud Recordings tab.

Services for Students with Disabilities and Academic Conduct

University of Michigan

If you think that you need an accommodation for a disability, please contact Services for Students with Disabilities (SSD) office to help us determine appropriate academic accommodations. SSD (734-763-3000; <http://ssd.umich.edu>) typically recommends accommodations through a Verified Individualized Services and Accommodations (VISA) form. Any information you provide is private and confidential and will be treated as such.

Clear definitions of the forms of academic misconduct, including cheating and plagiarism, as well as information about disciplinary sanctions for academic misconduct, may be found at the Rackham web site for the University of Michigan:

[http://www.rackham.umich.edu/policies/academic_policies/section10/.](http://www.rackham.umich.edu/policies/academic_policies/section10/)

Knowledge of these rules is the responsibility of the student and ignorance of them does not excuse misconduct. The student is expected to be familiar with these guidelines before submitting any written work or completing any projects in this course. Lack of familiarity with these rules in no way constitutes an excuse for acts of misconduct. Charges of plagiarism and other forms of academic misconduct will be dealt with very seriously and may result in oral or written reprimands, a lower or failing grade on the assignment, a lower or failing grade for the course, suspension, and/or, in some cases, expulsion from the university.

University of Maryland

For information about Disability Support Service (DSS) on campus, visit this web site:

<http://www.counseling.umd.edu/DSS/>

Disability Support Service (DSS) Main Office:

Phone: [301.314.7682](tel:301.314.7682)

Fax: [301.405.0813](tel:301.405.0813)

0106 Shoemaker Building

Dissup@umd.edu

Office Hours: Monday - Friday (8:30am to 4:30pm)

DSS Testing Office (for DSS exams):

Phone: [301.314.7217](tel:301.314.7217)

Fax: [301.314.9478](tel:301.314.9478)

0118 Shoemaker Building

DSSTest@umd.edu

DSS Exam Hours: Monday - Friday (9am to 4pm)

Overview of DSS Services:

In order to receive services you must contact the office to register in person for services. Please call the office to set up an appointment to register with a DSS counselor. Contact the DSS office at [301.314.7682](tel:301.314.7682).

There are a number of FAQs on the following page that you may find helpful:

http://www.counseling.umd.edu/DSS/add_questions.html

Clear definitions of the forms of academic misconduct, including cheating and plagiarism, as well as information about disciplinary sanctions for academic misconduct, may be found at the University of Maryland, Office of the President's website:

<http://www.president.umd.edu/policies/docs/III-100A.pdf>

Lecture Topics, Readings, and Schedule

August 25 — OVERVIEW; INTRODUCTION TO SURVEY ERRORS

No reading assignment is due but do give this week's readings a look if possible.

Required Readings:

Groves, R. M. (2004). *Survey Errors and Survey Costs*, Chapter 1, Sections 1.1-1.4 & 1.8

Biemer, P.P. and Lyberg, L.E. (2003). Introduction to Survey Quality, Chapter 2, Sections 2.1-2.5 (p 26-60)

Other key references:

Biemer, P. (2010). "Total Survey Error: Design, Implementation, and Evaluation." *Public Opinion Quarterly*, 74, pp. 817-848.

Groves, R. M. and Lyberg, Lars. (2010). "Total Survey Error: Past, Present, and Future." *Public Opinion Quarterly*, 74, 849-879.

Smith, T. W. (2011). Refining the total survey error perspective. *International Journal of Public Opinion Research*, 23(4), 464-484.

Lessler, J. and Kalsbeek, W. (1992). *Nonsampling Errors in Surveys*, John Wiley and Sons, Chapter 2.

Andersen, R. et al. (1979). *Total Survey Error*, New York: Academic Press.

Bailar, B. A. (1984). "The Quality of Survey Data." *1984 Proceedings of the Section on Survey Research Methods of the American Statistical Association*, pp 43-52.

U.S. Department of Commerce (1978). *A Glossary of Nonsampling Error Terms*. Washington, DC: Government Printing Office.

SEPTEMBER 1 – No class (Labor day)

SEPTEMBER 8 — COSTS, COVERAGE OF THE TARGET POPULATION

Reading Assignment #1 due Friday, September 5 by noon.

Required Readings:

Groves, R. M. (2004). *Survey Errors and Survey Costs*, Chapter 2

Groves, R. M. (2004). *Survey Errors and Survey Costs*, Chapter 3, Sections 3.1-3.4 (p 81- 99).*

Iannacchione, V. G. (2011). The changing role of address-based sampling in survey research. *Public Opinion Quarterly*, 75, 3, 556-575.

Other key references:

- Olson, K., Wagner, J. Anderson, R. Survey Costs: Where are We and What is the Way Forward?, *Journal of Survey Statistics and Methodology*, Volume 9, Issue 5, November 2021, Pages 921–942, <https://doi.org/10.1093/jssam/smaa014>
- Wagner, J., Guyer, H., Evanchek, C. Using Time Series Models to Understand Survey Costs, *Journal of Survey Statistics and Methodology*, Volume 9, Issue 5, November 2021, Pages 943–960, <https://doi.org/10.1093/jssam/smaa024>
- Blumberg, S. J. and Luke, J. V. (2009). Reevaluating the need for concern regarding noncoverage bias in landline surveys. *American Journal of Public Health*, 99(10), 1806-1810.
- Iannacchione, V., Staab, J. M., and Redden, D. T. (2003). "Evaluating the Use of Residential Mailing Addresses in a Metropolitan Household Survey," *Public Opinion Quarterly*, 67, 2, 202-210.
- Brick, J. M., Williams, D., and Montaquila, J. M. (2011). Address-based sampling for subpopulation surveys. *Public Opinion Quarterly*, 75(3), 409-428.
- Martin, E. 1999. Who knows who lives here: Within-household disagreements as a source of survey coverage error. *Public Opinion Quarterly* 63:220–36
- Mohorko, A., Leeuw, E. D., and Hox, J. (2013). Internet coverage and coverage bias in Europe: Developments across countries and over time. *Journal of official statistics*, 29, 609-622.
- Eckman, S. (2013). "Do Different Listers Make the Same Housing Unit Frame? Variability in Housing Unit Listing." *Journal of Official Statistics*, 29(2), 249-259.
- Fay, R.E. (1989). "An Analysis of Within-household Undercoverage in the Current Population Survey." *Proceedings of the U.S. Bureau of the Census Annual Research Conference* (pp. 156-175). Washington, DC: U.S. Bureau of the Census.
- Fein, D., and West, K. (1988). "Towards a Theory of Coverage Error: An Exploratory Assessment of Data from the 1986 Los Angeles Test Census." *Proceedings of the Fourth Annual Research Conference, Bureau of the Census* (pp. 540-562).
- Hogan, H. (1993). "The 1990 Post-Enumeration Survey: Operations and Results," *Journal of the American Statistical Association*, 88 (423), 1047-1060.
- Lessler, J. and Kalsbeek, W. (1992). *Nonsampling Errors in Surveys*, John Wiley and Sons, Chapters 3 and 4.
- Mulry, M. H. (2007). "Summary of accuracy and coverage evaluation for the U.S. Census 2000." *Journal of Official Statistics*, 23, 345-370.
- Robinson, J., Ahmed, B., das Gupta, P., and Woodrow, K., (1993). "Estimation of Population Coverage in the 1990 United States Census Based on Demographic Analysis," *Journal of the American Statistical Association*, 88 (423), 1061-1079.
- Brick, J. M., and Lepkowski, J. M.. (2008). "Multiple Mode and Frame Telephone Surveys." In J. M. Lepkowski, C. Tucker, J. M. Brick, E. D. de Leeuw, L. Japac, P. J. Lavrakas, M. W. Link, and R. L. Sangster (eds.), *Advances in Telephone Survey Methodology* (pp. 149-169). Hoboken, New Jersey: Wiley.
- Gaziano, C. (2005). Comparative Analysis of Within-Household Respondent Selection Techniques. *Public Opinion Quarterly*, 69:1, 124-157.

Keeter, S., Kennedy, C., Clark, A., Thompson, T., and Mokrzycki, M. (2007). “What’s Missing From National Landline RDD Surveys? The Impact of the Growing Cell-Only Population.” *Public Opinion Quarterly*, 71(5), 772–792.

Link, M. W., Battaglia, M. P., Frankel, M. R. Osborn, L., Mokdad A.H. (2007). “Reaching the U.S. Cell Phone Generation: Comparison of Cell Phone Survey Results with an Ongoing Landline Telephone Survey.” *Public Opinion Quarterly*, 71(5), 814–839.

Eckman, S. and Kreuter, F. (2011). “Confirmation Bias in Housing Unit Listing.” *Public Opinion Quarterly*, 75(1), 1-12.

Link, M.W., Battaglia, M.P., Frankel, M.R., Osborn, L. and Mokdad, A. (2008). “A Comparison of Address-Based Sampling (ABS) versus Random-digit Dialing (RDD) for General Population Surveys.” *Public Opinion Quarterly*, 72(1), 6-27.

Tourangeau, R., Shapiro, G., Kearney, A., and Ernst, L. (1997). “Who lives here? Survey undercoverage and household roster questions.” *Journal of Official Statistics*, 13, 1-18.

Tourangeau, R, Kreuter, F., and Eckman, S. (2012). “Motivated Underreporting in Screening Interviews.” *Public Opinion Quarterly*, 76(3), 453-469.

SEPTEMBER 15 – REPAIRS FOR UNDERCOVERAGE

Reading Assignment #2 due Friday, September 12 by noon.

Required Readings:

Groves, R. M. (2004). *Survey Errors and Survey Costs*, Chapter 3, Sections 3.6-3.8 (p 119-132).*

Harter, R., and English, N. (2018). Overview of three field methods for improving coverage of address-based samples for in-person interviews. *Journal of Survey Statistics and Methodology*, 6(3), 360-375.

Marpsat, M., and Razafindratsima, N. (2010). Survey methods for hard-to-reach populations: introduction to the special issue. *Methodological Innovations Online*, 5(2), 3-16.

Other key references:

Iachan, R. and Dennis, M. L. (1993). A multiple frame approach to sampling the homeless and transient population. *Journal of Official Statistics* 9, 747-764.

Kalton, G., Kali, J., and Sigman, R. (2014). Handling frame problems when address-based sampling is used for in-person household surveys. *Journal of Survey Statistics and Methodology*, 2(3), 283-304.

Eckman, S. (2016). Does the Inclusion of Non-Internet Households in a Web Panel Reduce Coverage Bias?. *Social Science Computer Review*, 34, 41-58.

Lessler, J. and Kalsbeek, W. (1992). Nonsampling Errors in Surveys, John Wiley and Sons, Chapter 5

Peytchev, A. and Neely, B. (2013). RDD telephone surveys: toward a single-frame cell-phone design. *Public Opinion Quarterly*, 77(1), 283-304.

Traugott, M. W., Groves, R. M., and Lepkowski, J. M. (1987). "Using Dual Frame Designs to Reduce Nonresponse in Telephone Surveys." *Public Opinion Quarterly* 51 (4):522-539.

Brick, J. M., and Lepkowski, J. M.. (2008). "Multiple Mode and Frame Telephone Surveys." In J. M. Lepkowski, C. Tucker, J. M. Brick, E. D. de Leeuw, L. Japac, P.J. Lavrakas, M. W. Link, and R. L. Sangster (eds.), *Advances in Telephone Survey Methodology* (pp. 149-169). Hoboken, New Jersey: Wiley

SEPTEMBER 22 – NONRESPONSE RATES AND NONRESPONSE ERROR

Reading Assignment #3 due Friday, September 19 by noon

Required Readings:

Groves, R. M. (2006). "Nonresponse rates and Nonresponse Bias in Household Surveys," *Public Opinion Quarterly*, 70:5, 646-675.*

Curtin, R., Presser, S., and Singer, E. (2000). "The Effects of Response Rate Changes on the Index of Consumer Sentiment." *Public Opinion Quarterly*, 64:4, 413-428.

Amaya, A., and Presser, S. (2016). "Nonresponse bias for univariate and multivariate estimates of social activities and roles." *Public opinion quarterly*, 81(1), 1-36.

Other key references:

Luiten, A., Hox, J., & de Leeuw, E. (2020). Survey Nonresponse Trends and Fieldwork Effort in the 21st Century: Results of an International Study across Countries and Surveys. *Journal of Official Statistics*, 36(3), 469-487.

Brick, J. M., and Williams, D. (2013). Explaining rising nonresponse rates in cross-sectional surveys. *The ANNALS of the American academy of political and social science*, 645(1), 36-59.

Keeter, S., Kohut, A., Miller, C., Groves, R., and Presser, S. (2000) . "Consequences of Reducing Nonresponse in a Large National Telephone Survey." *Public Opinion Quarterly*, 64:2, 125-148.

Groves, R. M. and Couper, M. P. (1998). *Nonresponse in household surveys*. New York: John Wiley, Chapter 1.

Groves, R., and Peytcheva, E. (2008). "The Impact of Nonresponse Rates on Nonresponse Bias: A Meta-Analysis." *Public Opinion Quarterly*, 72(2), 167-189.

Peytchev, A. (2013). "Consequences of Survey Nonresponse." *The ANNALS of the American Academy of Political and Social Science*, 645(1), 88-111.

Groves, R.M., Singer, E., and Corning, A. (2000). Leverage-Salience Theory of Survey Participation: Description and an Illustration. *Public Opinion Quarterly*, 64, 299-308.

De Leeuw, Edith, and de Heer, Wim. (2002). "Trends in Household Survey Nonresponse: A Longitudinal and International Comparison." In R. Groves, D. Dillman, J. Eltinge, and R. Little (eds.) *Survey Nonresponse* (pp. 41-54). New York: Wiley.

- Groves, R. M., and Couper, M. P. (1998). *Nonresponse in household surveys*. New York: John Wiley, Chapter 2.
- Abraham, K. G., Maitland, A., and Bianchi, S. M. (2006). "Nonresponse in the American Time Use Survey: Who is Missing from the Data and How Much Does it Matter?" *Public Opinion Quarterly*, 70:5, 676-703.
- Keusch, F. (2015). Why do people participate in Web surveys? Applying survey participation theory to Internet survey data collection. *Management review quarterly*, 65(3), 183-216.
- Fan, W., & Yan, Z. (2010). Factors affecting response rates of the web survey: A systematic review. *Computers in human behavior*, 26(2), 132-139.
- Tourangeau, R. (2017). Presidential Address: Paradoxes of Nonresponse. *Public Opinion Quarterly*.
- DeMaio, T. J. (1980). "Refusals: Who, Where, and Why?" *Public Opinion Quarterly*, 223-233.
- Groves, R., Cialdini, R., and Couper, M. P. (1992). "Understanding the Decision to Participate in a Survey," *Public Opinion Quarterly*, 56:4, 475-495.
- Groves, R. M., Presser, S., and Dipko, S. (2004). The role of topic interest in survey participation decisions. *Public Opinion Quarterly*, 68:1, 2-31.
- Kreuter, F., Mueller, G., Trappmann, M. (2010). "Nonresponse and Measurement Error in Employment Research." *Public Opinion Quarterly*, 74(5), 880-906.
- Merkle, D., and Edelman, M. (2002). "Nonresponse in exit polls: A comprehensive analysis." In R. Groves, D. Dillman, J. Eltinge, & R. Little (Eds.), *Survey Nonresponse* (pp. 243-258). New York: John Wiley.
- Teitler, J. O., Reichman, N. E., and Sprachman, S. (2003). "Costs and benefits of improving response rates for Hard-to-Reach Population." *Public Opinion Quarterly*, 67:1, 126-138.
- Lepkowski, James M., and Mick P. Couper. 2002. "Nonresponse in the Second Wave of Longitudinal Household Surveys." In *Survey Nonresponse*, edited by R. Groves, D. Dillman, J. Eltinge and R.J.A. Little, 259-272. New York: Wiley.
- Singer, E., Mathiowetz, N., and Couper, M. (1993). "Privacy, Confidentiality, and the 1990 U.S. Census." *Public Opinion Quarterly*, 57, 465-483.

September 29 – DEALING WITH NONRESPONSE

Reading Assignment #4 due Friday, September 26 by noon

Required Readings:

- Lessler, J. and Kalsbeek, W. (1992). *Nonsampling Errors in Surveys*, John Wiley and Sons, Chapter 8, Sections 8.0 - 8.1.5.
- Groves, R.M. et al. 2008. Issues Facing the Field: Alternative Practical Measures of Representativeness of Survey Respondent Pools. *Survey Practice*, 1 (3).
- Brick, J. M., and Tourangeau, R. (2017). Responsive survey designs for reducing nonresponse bias. *Journal of Official Statistics*, 33(3), 735-752.*

Other Key References:

Tourangeau, R., Michael Brick, J., Lohr, S., and Li, J. (2017). Adaptive and responsive survey designs: a review and assessment. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 180(1), 203-223.

Singer, E. (2011). "Toward a Benefit-Cost Theory of Survey Participation: Evidence, Further Tests, and Implications." *Journal of Official Statistics*, 27(2), 379-392.

Peytchev, A., Baxter, R. K., and Carley-Baxter, L. R. (2009). Not all survey effort is equal: Reduction of nonresponse bias and nonresponse error. *Public Opinion Quarterly*, 73(4), 785-806.

Groves, R. M., Couper, M. P., Presser, S., Singer, E., Tourangeau, R., Acosta, G. P., and Nelson, L. (2006). Experiments in producing nonresponse bias. *International Journal of Public Opinion Quarterly*, 70(5), 720-736.

Lin, I-Fen, Schaeffer, N. "Using Survey Participants to Estimate the Impact of Nonparticipation." *Public Opinion Quarterly* 1995;59:236-58.

Iannacchione, V. G. 2003. "Sequential weight adjustments for location and cooperation propensity for the 1995 National Survey of Family Growth." *Journal of Official Statistics*, 19:31-43.

Schouten, B., Shlomo, N., and Skinner, C. (2011) "Indicators for Monitoring and Improving Representativeness of Response." *Journal of Official Statistics*, 27(2), 231-253.

Lessler, J. and Kalsbeek, W. (1992). *Nonsampling Errors in Surveys*, John Wiley and Sons, Chapter 8, Sections 8.0 - 8.1.6.

Mercer, A., Caporaso, A., Cantor, D., & Townsend, R. (2015). How much gets you how much? Monetary incentives and response rates in household surveys. *Public Opinion Quarterly*, 79(1), 105-129.

Andridge, R. and Little, R. A. (2011) "Proxy Pattern-Mixture Analysis for Survey Nonresponse." *Journal of Official Statistics*, 27:2, 153-180.

West, Brady T., and Robert M. Groves. (2013). "A Propensity-Adjusted Interviewer Performance Indicator." *Public Opinion Quarterly*, 77, 1, 352-374.

Wagner, J. (2012). A comparison of alternative indicators for the risk of nonresponse bias. *Public Opinion Quarterly*, 76(3), 555-575.

O'Muircheartaigh, C., and Campanelli, P. (1999). "A multilevel exploration of the role of interviewers in survey non-response." *Journal of the Royal Statistical Association, Series A*, 162, Part 3, 437-446.

Couper, M. P., Singer, E., Conrad, F. G., & Groves, R. M. (2008). Risk of disclosure, perceptions of risk, and concerns about privacy and confidentiality as factors in survey participation. *Journal of official statistics*, 24(2), 255.

OCTOBER 6 – WEIGHTING AND IMPUTATION + ANALYTICAL ERROR.

Reading Assignment #5 due Friday, October 3 by noon

Required Readings:

Brick, J. M., and Kalton, G. (1996). Handling missing data in survey research. *Statistical Methods in Medical Research*, 5, 215 – 238.

Little, R. J., and Vartivarian, S. L. (2005). "Does Weighting for Nonresponse Increase the Variance of Survey Means?" *Survey Methodology*, 31, 161-168.

Kreuter, F., Olson, K., Wagner, J., Yan, T., Ezzati-Rice, T. M., Casas-Cordero, C., ... and Raghunathan, T. E. (2010). Using proxy measures and other correlates of survey outcomes to adjust for non-response: examples from multiple surveys. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 173(2), 389-407.*

West, B. T., Sakshaug, J. W., & Aurelien, G. A. S. (2016). How big of a problem is analytic error in secondary analyses of survey data?. *PLoS One*, 11(6), e0158120.

Other Key References:

Tourangeau, R. Conrad, F., & Couper M. 2013. "Chapter 2: Sampling and Coverage Issues for Web Surveys," *The Science of Web Surveys*. 11-35.

Cohn, N. (2016). "We Gave Four Good Pollsters the Same Raw Data. They Had Four Different Results." *The New York Times*. Available here: <https://www.nytimes.com/interactive/2016/09/20/upshot/the-error-the-polling-world-rarely-talks-about.html?module=inline>

Bethlehem, J. G. (2002). "Weighting Nonresponse Adjustments Based on Auxiliary Information." In Groves, R., Dillman, D., Eltinge, J., and Little, R. (eds.) *Survey Nonresponse* (pp. 275-288). New York: Wiley, 2002.

Dever, J.A., Rafferty, A., and Valliant, R. (2008). "Internet Surveys: Can Statistical Adjustments Eliminate Coverage Bias?" *Survey Research Methods*, 2(2), 47-62.

Rubin, D. (1986). "Basic Ideas of Multiple Imputation for Nonresponse," *Survey Methodology*, 12(1), 37-47.

Pew Research Center, January, 2018, "For Weighting Online Opt-In Samples, What Matters Most?"

Schenker, N., Trivellore Raghunathan, P.L. Chiu, D.M. Makuc, G.Y. Zhang, and A.J. Cohen. 2006. "Multiple imputation of missing income data in the National Health Interview Survey." *Journal of the American Statistical Association*, 101(475): 924-933.

Valliant, Richard, Jill A. Dever, and Frauke Kreuter. (2013). *Practical Tools for Designing and Weighting Survey Samples*, Chapter 13, Sections 13.5 – 13.8. New York, NY: Springer.

Lee, S. & Valliant, R. (2008). "Weighting Telephone Samples using Propensity Scores." In J. M. Lepkowski, C. Tucker, J. M. Brick, E. D. de Leeuw, L. Japac, P. J. Lavrakas, M. W. Link, and R. L. Sangster (eds.), *Advances in Telephone Survey Methodology* (pp. 170-183). Hoboken, New Jersey: Wiley.

Kreuter, F. and Olson, K. (2011). "Multiple Auxiliary Variables in Nonresponse Adjustment." *Sociological Methods and Research*, 40(2), 311-332.

Binder, D., Michaud, S., and Poirer, C. (1994). "Model-based reweighting for nonresponse adjustment." Paper presented at the COPAFS Symposium on New Directions in Statistical Methodology, May.

Chapman, D., Bailey, L., and Kasprzyk, D. (1986). "Nonresponse Adjustment Procedures at the U.S. Bureau of the Census." *Survey Methodology*, 12:1, 161-180.

Holt, D. and Elliot, D. (1991), "Methods of weighting for unit non-response." *The Statistician*, 40: 333-342.

Holt, D., and Smith, T.M.F. (1979). "Post Stratification," *Journal of the Royal Statistical Society, Series A*, 142:1, 33-46.

Kalton, Graham, and Flores-Cervantes, Ismael (2003). "Weighting Methods." *Journal of Official Statistics*, 19:2, 81-97.

Kish, L. (1992). "Weighting for Unequal P_i ," *Journal of Official Statistics*, 8:2, 183-200.

Little, R., and Rubin, D. (2002). *Statistical Analysis with Missing Data, Second Edition*. New York: Wiley.

Marker, D. A., Judkins, D. R., and Winglee, M. (2001). "Large-scale Imputation for Complex Surveys," In Groves, R., Dillman, D., Eltinge, J., and Little, R. (eds.) *Survey Nonresponse* (pp. 329-342). New York: Wiley.

West, B.T. (2013). "The Effects of Error in Paradata on Weighting Class Adjustments: A Simulation Study." *Chapter 15 in Improving Surveys with Paradata: Analytic Use of Process Information*. Wiley Publishing.

West, B.T. and Little, R.J.A. (2013). "Nonresponse Adjustment of Survey Estimates Based on Auxiliary Variables Subject to Error." *Journal of the Royal Statistical Society, Series C (Applied Statistics)*, 62(2), 213-231

Ragunathan, T. E. and J. E. Grizzle. 1995 "A split questionnaire survey design" *Journal of American Statistical Association*, 90:55-63.

Kalton, G., and Kasprzyk, D. (1986). "The treatment of missing survey data." *Survey Methodology*, 12:1, 1-16.

OCTOBER 13 – NO CLASS. Fall study break.

OCTOBER 20 – MIDTERM EXAMINATION

OCTOBER 27 – MEASUREMENT ERROR OVERVIEW

Reading Assignment #6 due Friday, October 24 by noon

Required Readings:

Groves, Robert M. (2004). *Survey Errors and Survey Costs*, Chapter 1, Section 1.5

Ghiselli, B. E., Campbell, J. P., & Zedeck, S. (1981). Measurement theory for the behavioral sciences. Chapter 10 – pp. 265 - 267 (ending at "Criterion-related Validity") and 280 (starting with "Construct Validity") - 287.

Bohrnstedt, George W. (1983). "Measurement." In Rossi, P. H., Wright, J. D., and Anderson, A. B. (eds.), *Handbook of Survey Research*. New York: Academic Press. Sections 3.0-3.5 (pp. 69-79)*

Other Key References:

Miller, Peter V. (2011). "Measurement Error." In Lavrakas, P. (eds.) *Encyclopedia of Survey Research Methods* (pp. 458-461). Thousand Oaks: Sage.

Ghiselli, B. E., Campbell, J. P., & Zedeck, S. (1981). *Measurement theory for the behavioral sciences*. Chapter 10 – pp. 265- 287.

Groves, R. M. (1991). Measurement error across disciplines. In Biemer, P. P., Groves, R. M., Lyberg, L. E., Mathiowetz, N. A., & Sudman, *Measurement errors in surveys*. Sections 1.1-1.2 (p 1-9).

Biemer, P. and Stokes, L. (1991). "Approaches to the Modeling of Measurement Errors," in P. Biemer *et al.* (eds.) *Measurement Errors in Surveys*, New York: Wiley, 1991, pp. 487-516.

Hansen, M. H., Hurwitz, H. and Bershad, M. (1960). "Measurement Errors in Censuses and Surveys." *Bulletin of the International Statistical Institute*, 32nd Session, 38(2), 359-374.

Biemer, P., and Trewin, D. (1997). "A Review of Measurement Error Effects on the Analysis of Survey Data." In Lyberg, L., Biemer, P., Collins, M., de Leeuw, E., Dippo, C., Schwarz, N., and Trewin, D. (eds.), *Survey Measurement and Process Quality* (pp. 603-632). New York: Wiley.

Fuller, W. *Measurement Error Models*. New York: Wiley, 1987. Chapter 1, Section 1.1.

Lessler, J. T. (1985). "Measurement Error in Surveys." In Turner, C. F., and Martin, E. (eds.), *Surveying Subjective Phenomena* (Volume 2, pp. 405-440). New York: Russell Sage Foundation.

Lessler and Kalsbeek, Chapter 10

Turner, C. F., and Martin, E. (1985). "Measurement and Error: An Introduction." In Turner, C. F., and Martin, E. (eds.), *Surveying Subjective Phenomena* (Volume 1, pp. 97-128). New York: Russell Sage Foundation.

NOVEMBER 3 – METHODS OF STUDYING MEASUREMENT ERROR

Reading Assignment #7 due Friday, October 31 by noon

Required Readings:

Groves, R. M. (2004). *Survey Errors and Survey Costs*, Chapter 7.

Parry, H. J. & Crossley, H. M. (1950). Validity of responses to survey questions. *Public Opinion Quarterly*, 14(1), 61-80.*

Other Key References:

Biemer, P.P. and Lyberg, L.E. (2003). Overview of survey error evaluation methods. Chapter 8 in *Introduction to Survey Quality*, New York: Wiley, 258-304.

Maitland, A. & Presser, S. (2016). How accurately do different evaluation methods predict the reliability of survey questions?. *Journal of Survey Statistics and Methodology*, 4(3), 362-381.

Antoun, C., Couper, M. P., & Conrad, F. G. (2017). Effects of mobile versus PC Web on survey response quality: a crossover experiment in a probability web panel. *Public Opinion Quarterly*, 81(S1), 280-306.

Alwin, D. (2007). *Margins of Error*. New York: Wiley.

Kreuter, F., Presser, S., & Tourangeau, R. (2008). Social desirability bias in CATI, IVR, and web surveys: The effects of mode and question sensitivity. *Public Opinion Quarterly*, 72(5), 847-865.

Biemer, P. (2004). "Modeling Measurement Error to Identify Flawed Questions." In Presser, S., Couper, M. P., Lessler, J. T., Martin, E., Martin, J., Rothgeb, J. M., and Singer, E. (eds.) *Methods for Testing and Evaluation Survey Questionnaires* (pp. 225-246). New York: Wiley.

Andrews, F.M. (1984). "Construct Validity and Error Components of Survey Measures: A Structural Modeling Approach," *Public Opinion Quarterly*, 48:2, 409-442.

Kreuter, F., Yan, T., and Tourangeau, R. (2008). "Good item or bad-can latent class analysis tell? The utility of latent class analysis for the evaluation of survey questions". *Journal of the Royal Statistical Society*, 171(3), 723-738.

Presser, S., Couper, M. P., Lessler, J. T., Martin, E., Martin, J., Rothgeb, J. M., & Singer, E. (2004). Methods for testing and evaluating survey questions. *Public Opinion Quarterly*, 68(1), 109-130.

West, Brady T., and Olson, Kristen (2010). "How much of interviewer variance is really nonresponse error variance?" *Public Opinion Quarterly*, 74, 1004-1026.

Forsyth, B.H. and Lessler, J.T. 1991. "Cognitive Laboratory Methods: A Taxonomy" in P. Biemer, R.M. Groves, L.E. Lyberg, N.A. Mathiowetz, and S. Sudman, (eds), *Measurement Errors in Surveys*. New York: Wiley, pp. 393-418.

Saris, W., van Wijk, T, and Scherpenzeel, A., (1998) "Validity and Reliability of Subjective Social Indicators: the Effect of Different Measures of Association," *Social Indicators Research*, 45:1-3, 173-199.

Elliott, M. N., Zaslavsky, A. M., Goldstein, E., Lehrman, W., Hambarsoomians, K., Beckett, M. K., & Giordano, L. (2009). Effects of Survey Mode, Patient Mix, and Nonresponse on CAHPS® Hospital Survey Scores. *Health Services Research*, 44(2p1), 501-518.

O'Muircheartaigh, C. (1991). "Simple Response Variance: Estimation and Determinants." In Biemer, P., Groves, R. M., Lyberg, L., Mathiowetz, N., and Sudman, S. (eds.), *Measurement Errors in Surveys* (pp. 551-574). New York: Wiley.

Saris, W., and Andrews, F. (1991). "Evaluation of Measurement Instruments Using a Structural Modeling Approach." In Biemer, P., Groves, R. M., Lyberg, L., Mathiowetz, N., and Sudman, S. (eds.), *Measurement Errors in Surveys* (pp. 575-598). New York: Wiley.

Oberski, D. (2012). ["Comparability of survey measurements."](#) In *Handbook of Survey Methodology in Social Sciences* (L. Gideon, ed.), Springer, New York.

Saris, W., van Wijk, T, and Scherpenzeel, A., (1998) "Validity and Reliability of Subjective Social Indicators: the Effect of Different Measures of Association," *Social Indicators Research*, 45:1-3, 173-199.

NOVEMBER 10 – SOURCES OF MEASUREMENT ERROR: THE ROLE OF THE INTERVIEWER, RESPONDENT, AND QUESTIONNAIRE

Reading Assignment #8 due Friday, November 7 by noon

Required Readings:

Federal Committee on Statistical Methodology. (2001). Measuring and Reporting Sources of Error in Surveys. Chapter 6, Sections 6.1-6.2 (p 1-12).

West, B. T. & Blom, A. G. (2016). Explaining interviewer effects: A research synthesis. *Journal of Survey Statistics and Methodology*, 5(2), 175-211.*

Groves, R. M. (2004). *Survey Errors and Survey Costs*, Chapter 9.

Other Key References:

Krosnick, J. A. (1991). Response strategies for coping with the cognitive demands of attitude measures in surveys. *Applied cognitive psychology*, 5(3), 213-236.

Schaeffer, N.C., and Presser, S. (2003). "The Science of Asking Questions." *Annual Review of Sociology*, 29, pp. 65-88.

Krosnick, J. A., & Presser, S. (2010). Question and questionnaire design. *Handbook of survey research*, 2(3), 263-314.

Conrad, F.G., & Schober, M.F. (2000). Clarifying question meaning in a household telephone survey. *Public Opinion Quarterly*, 64, 1-28.

O'Muircheartaigh, C., and Campanelli P. (1998). "The relative impact of interviewer effects and sample design effects on survey precision." *Journal Of The Royal Statistical Society, Series A*, 161:1, 63-77.

Cannell, C. F., Miller, P. V., and Oksenberg, L. (1981). "Research on Interviewing Techniques." In S. Leinhardt (ed.), *Sociological Methodology, 1981* (pp. 389-447). San Francisco: Jossey-Bass.

Fowler, F. J., Jr. and Mangione, T. W. (1985). "The Value of Interviewer Training and Supervision." Boston, MA: Center for Survey Research.

Groves, R. M. and L. Magilavy (1980). "Estimates of Interviewer Variance for Telephone Surveys." *Proceedings of the Survey Research Methods Section of the American Statistical Association*.

Hatchett, S., and Schuman, H. (1975). "Race of Interviewer Effects Upon White Respondents." *Public Opinion Quarterly*, 39:4, 523-527.

O'Muircheartaigh, C.A. (1977). "Response Errors." In O'Muircheartaigh, C. A., and Payne, C. (eds.), *The Analysis of Survey Data*, Volume 2. New York: Wiley.

Schuman, H., and Converse, J. M. (1971). "The Effects of Black and White Interviewers on Black Responses in 1968," *Public Opinion Quarterly*, 35:1, 44-68.

Singer, E., and Kohnke-Aguirre, L. (1979). "Interviewer Expectation Effects: A Replication and Extension," *Public Opinion Quarterly*, 1979, Vol. 43, No. 2, pp. 245-260.

Tourangeau, R., and Bradburn, N. M. (2010). "The Psychology of Survey Response." In P.V. Marsden and J.D. Wright (Eds.), *The Handbook of Survey Research*, Second Edition (pp. 315-346). Bingley, UK: Emerald.

Gaskell, G., Wright, D. and O'Muircheartaigh, C. (2000). Telescoping landmark events: Implications for survey research. *Public Opinion Quarterly*, 64, 77-89.

- Rips, L. J., Conrad, F. G., and Fricker, S. S., (2003). "Straightening the Seam Effect in Panel Surveys." *Public Opinion Quarterly*, 67:4, 522-554.
- Kalton, G., and Miller, M. E. (1991). The seam effect with social security income in the Survey of Income and Program Participation. *Journal of Official Statistics*, 7(2), 235-245
- Gaskell, G., Wright, D. and O'Muircheartaigh, C. (2000). Telescoping landmark events: Implications for survey research. *Public Opinion Quarterly*, 64, 77-89.
- Tourangeau, R., and Smith, T. W. 1996. "Asking sensitive questions - The impact of data collection mode, question format, and question context." *Public Opinion Quarterly* 60 (2):275-304.
- Neter, J., and Waksberg, J. 1964. "A Study of Response Errors in Expenditures Data from Household Interviews." *Journal of the American Statistical Association* 59:18-55.
- Schuman, H., and Presser, S. 1981. *Questions and Answers in Attitude Surveys: Experiments on Question Form, Wording, and Context, Quantitative studies in social relations*. New York: Academic Press. Chapter 11.

NOVEMBER 17 – PROCESSING ISSUES IN SURVEYS

Reading Assignment #9 due Friday, November 14 by noon

Required Readings:

- Federal Committee on Statistical Methodology. (2001). Measuring and Reporting Sources of Error in Surveys. Chapter 7.*
- Lupia, A. (2018). Coding open responses. In *The Palgrave Handbook of Survey Research* (pp. 473-487). Palgrave Macmillan, Cham.
- Reiter, J. P. (2012). "Research Synthesis: Statistical Approaches To Protecting Confidentiality For Microdata And Their Effects On The Quality Of Statistical Inferences." *Public Opinion Quarterly*, 76(1): 163-181.

Other Key References:

- De Waal, T., Pannekoek, J., and Scholtus, S. (2011). *Handbook of statistical data editing and imputation* (Vol. 563). John Wiley & Sons. Chapter 1.
- Schonlau, M. and Couper, M. P. (2016). Semi-automated categorization of open-ended questions. *Survey Research Methods*, 10(2), 143-152.
- Hilton, K. (2016). "Introduction to Data Linkage". Available here: https://adrm.ac.uk/media/174200/data_linkage_katieharron_2016.pdf
- Winkler, W. E. (2014). Matching and record linkage. *Wiley Interdisciplinary Reviews: Computational Statistics*, 6(5), 313-325.

Granquist, L., and Kovar, J. (1997). "Editing of Survey Data: How Much is Enough." In Lyberg, L., Biemer, P., Collins, M., de Leeuw, E., Dippo, C., Schwarz, N., and Trewin, D. (eds.), *Survey Measurement and Process Quality* (pp. 415-436). New York: Wiley.

Campanelli, P., Thomson, K., Moon, N., and Staples, T. (1997). "The Quality of Occupational Coding in the United Kingdom." In Lyberg, L., Biemer, P., Collins, M., de Leeuw, E., Dippo, C., Schwarz, N., and Trewin, D. (eds.), *Survey Measurement and Process Quality* (pp. 437-456). New York: Wiley.

Winkler, W.E. (1995). "Matching and Record Linkage." In Cox, B., Binder, D. A., Chinnappa, B. N., Christianson, A., Colledge, M.J., and Kott, P. (eds.), *Business Survey Methods* (pp. 355-384). New York: Wiley.

Fellegi, I., and Holt, D. (1976). "A Systematic Approach to Automatic Edit and Imputation," *Journal of the American Statistical Association*, 71:353, 17-35.

Biemer, P., and Lyberg, L. (2004). *Introduction to Survey Quality*, Chapter 7. New York: Wiley.

Hidioglou, M., and Berthelot, J. (1986). "Statistical Editing and Imputation for Periodic Business Surveys." *Survey Methodology*, 12:1, 73-83.

Little, R., and Smith, P. (1987). "Editing and Imputation for Quantitative Survey Data." *Journal of the American Statistical Association*, 82:397, 58-67.

Morganstein, D., and Marker, D. (1997). "Continuous Quality Improvement in Statistical Agencies." In Lyberg, L., Biemer, P., Collins, M., de Leeuw, E., Dippo, C., Schwarz, N., and Trewin, D. (eds.), *Survey Measurement and Process Quality* (pp. 475-500). New York: Wiley.

Pierzchala, M., "Editing Systems and Software." In Cox, B., Binder, D. A., Chinnappa, B. N., Christianson, A., Colledge, M.J., and Kott, P. (eds.), *Business Survey Methods* (pp. 425-441). New York: Wiley.

Pierzchala, M. (1990). "A Review of the State of the Art in Automated Data Editing and Imputation." *Journal of Official Statistics*, 6:4, 355-377.

Statistical Policy Office, Office of Management and Budget. (1990). *Data Editing in Federal Statistical Agencies*. Washington, DC: Office of Management and Budget.

Rasner, A. Frick J. R., Grabka, M. (2013). "Statistical Matching of Administrative and Survey Data: An Application to Wealth Inequality Analysis," *Sociological Methods & Research*, 42, 192-224.

NOVEMBER 24 – MULTIPLE SOURCES OF SURVEY ERROR

Reading Assignment #10 due Friday, November 21 by noon

Required Readings:

Biemer, P.P. and Lyberg, L.E. (2003). *Introduction to Survey Quality*, Chapter 2, Sections 2.6 (p 60-62)

Tourangeau, R. (2020). How Errors Cumulate: Two Examples. *Journal of Survey Statistics and Methodology*, 8(3), 413-432.*

Olson, Kristen. (2006). "Survey participation, nonresponse bias, measurement error bias, and total bias." *Public Opinion Quarterly* 70 (5):737-758.

Other Key References:

Federal Committee on Statistical Methodology. (2001). Measuring and Reporting Sources of Error in Surveys. Chapter 8.

Antoun, C., Conrad, F. G., Couper, M. P., & West, B. T. (2019). Simultaneous estimation of multiple sources of error in a smartphone-based survey. *Journal of Survey Statistics and Methodology*, 7(1), 93-117.

Tourangeau, R., Groves, R. M., and Redline, C. D. (2010). Sensitive topics and reluctant respondents: Demonstrating a link between nonresponse bias and measurement error. *Public Opinion Quarterly*, 74(3), 413-432.

Shirani-Mehr, H., Rothschild, D., Goel, S., and Gelman, A. (2018). Disentangling bias and variance in election polls. *Journal of the American Statistical Association*, 113(522), 607-614.

Kennedy, C., Blumenthal, M., Clement, S., Clinton, J. D., Durand, C., Franklin, C., ... and Saad, L. (2018). An evaluation of the 2016 election polls in the United States. *Public Opinion Quarterly*, 82(1), 1-33.

Sakshaug, J. W., Yan, T., & Tourangeau, R. (2010). Nonresponse error, measurement error, and mode of data collection: Tradeoffs in a multi-mode survey of sensitive and non-sensitive items. *Public Opinion Quarterly*, 74(5), 907-933.

Sturgis, P., Kuha, J., Baker, N., Callegaro, M., Fisher, S., Green, J., ... & Smith, P. (2018). An assessment of the causes of the errors in the 2015 UK general election opinion polls. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 181(3), 757-781.

Felderer, B., Kirchner, A., & Kreuter, F. (2019). The Effect of Survey Mode on Data Quality: Disentangling Nonresponse and Measurement Error Bias. *Journal of Official Statistics*, 35(1), 93-115.

Peytchev, Andy, Emilia Peytcheva, and Robert M. Groves. (2010). "Measurement Error, Unit Nonresponse, and Self-Reports of Abortion Experiences." *Public Opinion Quarterly* 74 (2):319-327.

Groves, Robert M. and Lou J. Magilavy. (1984) "An Experimental Measurement of Total Survey Error," *Proceedings of the Survey Research Methods Section of the American Statistical Association*, 1984.

Cannell, C. F., and Fowler, F. J. 1963. "Comparison of a Self-Enumerative Procedure and a Personal Interview: A Validity Study." *Public Opinion Quarterly* 27:250-264.

Peytchev, A., Carley-Baxter, L A., Black, M C. (2011). "Multiple Sources of Nonobservation Error in Telephone Surveys: Coverage and Nonresponse." *Sociological Methods & Research*.

Eckman, S., & Kreuter, F. (2017). The Undercoverage-Nonresponse Trade-Off. *Total Survey Error in Practice*, 95-113.

Voogt, R. J. J., & van Kempen, H. (2002). Nonresponse Bias and Stimulus Effects in the Dutch National Election Study. *Quality and Quantity*, 36(4), 325-345.

Peytchev, Andy, Jamie Ridenhour and Karol Krotki (2010). Telephone and ABS Mail Survey Design: Coverage, Unit Nonresponse, and Measurement Error. *Journal of Health Communication: International Perspectives* 15(3): 117-34.

Tourangeau, R., Groves, R. M., Kennedy, C. and Yan, T. (2009) The presentation of a web survey, nonresponse and measurement error among members of web panel. *Journal of Official Statistics*, 25, 299–321.

DECEMBER 1 – TSE AND FOUND DATA. Emerging data collection methods and large-language models.

No reading assignment is due but do read the papers below so we can have a good discussion in class.

Callegaro, M. and Yang, Y. (2018). The role of surveys in the era of “big data”. In *The Palgrave handbook of survey research* (pp. 175-192). Palgrave Macmillan, Cham.

Amaya, A., Biemer, P. P., & Kinyon, D. (2020). Total Error in a Big Data World: Adapting the TSE Framework to Big Data. *Journal of Survey Statistics and Methodology*, 8(1), 89-119.

Sen, I., Flöck, F., Weller, K., Weiß, B., & Wagner, C. (2021). A total error framework for digital traces of human behavior on online platforms. *Public Opinion Quarterly*, 85(S1), 399-422.

Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B., & Larson, J. M. (2023). Synthetic replacements for human survey data? The perils of large language models. *Political Analysis*, 1-16.

Other Key References:

Wang, W., Rothschild, D., Goel, S., & Gelman, A. (2015). Forecasting elections with non-representative polls. *International Journal of Forecasting*, 31(3), 980-991.

Salganik, M. J. (2017). *Bit by bit: social research in the digital age*. Princeton University Press. Chapters 2-3. Available here: <https://www.bitbybitbook.com/en/1st-ed/preface/>

Hill, C. A., Biemer, P., Buskirk, T., Callegaro, M., Cazar, A. L. C., Eck, A., ... and Sturgis, P. (2019, April). Exploring New Statistical Frontiers at the Intersection of Survey Science and Big Data: Convergence at “BigSurv18”. In *Survey Research Methods* (Vol. 13, No. 1, pp. 123-134).

Keusch, F., Bähr, S., Haas, G. C., Kreuter, F., & Trappmann, M. (2023). Coverage error in data collection combining mobile surveys with passive measurement using apps: Data from a German national survey. *Sociological Methods & Research*, 52(2), 841-878.

Rocha-Silva, T., Nogueira, C., & Rodrigues, L. (2024). Passive data collection on Reddit: a practical approach. *Research Ethics*, 20(3), 453-470.

Biemer, P. (2016). “Chapter 10: Errors and Inference.” In I. Foster et al. (eds). *Big data and social science: A practical guide to methods and tools*. Chapman and Hall/CRC.

Japac, L., Kreuter, F., Berg, M., Biemer, P., Decker, P., Lampe, C., ... and Usher, A. (2015). Big data in survey research: AAPOR task force report. *Public Opinion Quarterly*, 79 (4), 839-880.

Groves, R. M. (2011). Three eras of survey research. *Public Opinion Quarterly*, 75(5), 861-871.

DECEMBER 8 -- FINAL EXAMINATION